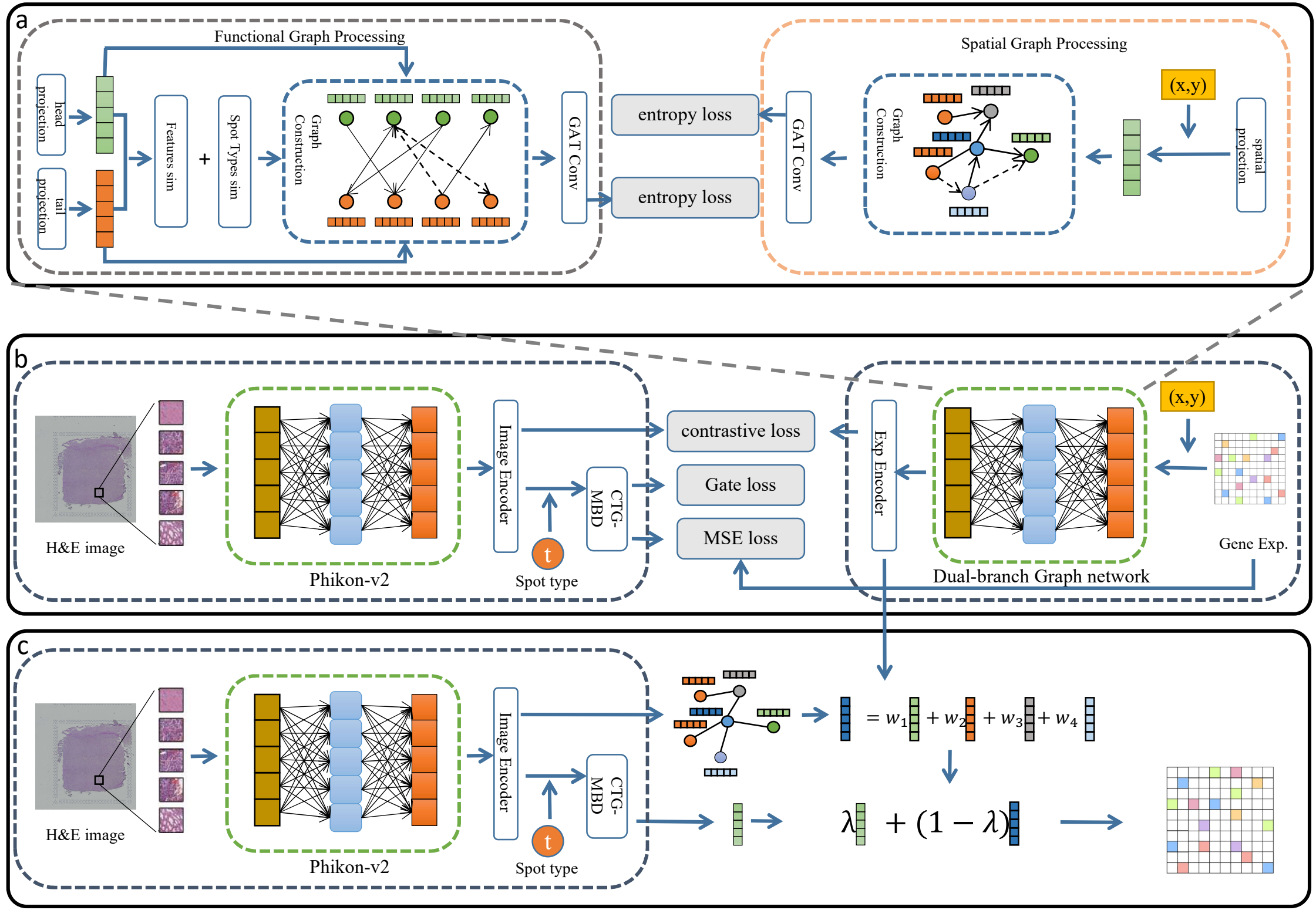
a
Functional Graph Processing
head projection
tail projection
Features sim
+
Spot Types sim
Graph Construction
GAT Conv
entropy loss
entropy loss
Spatial Graph Processing
GAT Conv
Graph Construction
(x,y)
spatial projection
b
H&E image
Phikon-v2
Image Encoder
CTG-MBD
t
Spot type
contrastive loss
Gate loss
MSE loss
Exp Encoder
(x,y)
Gene Exp.
Dual-branch Graph network
c
H&E image
Phikon-v2
Image Encoder
CTG-MBD
t
Spot type
$= w_1 + w_2 + w_3 + w_4$
$\lambda + (1-\lambda)$